

Python SQL Library Comparison — Core Objects

Connection & Cursor reference across PostgreSQL, SQLite, MySQL, Oracle, and Microsoft SQL Server

Feature	PostgreSQL psycopg3	SQLite sqlite3	MySQL mysql-connector-python	Oracle cx_Oracle	MS SQL Server pyodbc
Install	pip install psycopg[binary]	Built-in (no install needed)	pip install mysql-connector-python	pip install cx_Oracle	pip install pyodbc
Connect	psycopg.connect()	sqlite3.connect()	mysql.connector.connect()	cx_Oracle.connect()	pyodbc.connect()
Connection Object	psycopg.Connection	sqlite3.Connection	CMySQLConnection	cx_Oracle.Connection	pyodbc.Connection
Cursor Object	psycopg.Cursor	sqlite3.Cursor	CMySQLCursor	cx_Oracle.Cursor	pyodbc.Cursor
Create Cursor	conn.cursor()	conn.cursor()	conn.cursor()	conn.cursor()	conn.cursor()
Execute SQL	cursor.execute()	cursor.execute()	cursor.execute()	cursor.execute()	cursor.execute()
Fetch One	cursor.fetchone()	cursor.fetchone()	cursor.fetchone()	cursor.fetchone()	cursor.fetchone()
Fetch All	cursor.fetchall()	cursor.fetchall()	cursor.fetchall()	cursor.fetchall()	cursor.fetchall()
Fetch Many	cursor.fetchmany(n)	cursor.fetchmany(n)	cursor.fetchmany(n)	cursor.fetchmany(n)	cursor.fetchmany(n)
Commit	conn.commit()	conn.commit()	conn.commit()	conn.commit()	conn.commit()
Rollback	conn.rollback()	conn.rollback()	conn.rollback()	conn.rollback()	conn.rollback()
Close Connection	conn.close()	conn.close()	conn.close()	conn.close()	conn.close()
Context Manager	with psycopg .connect()	with sqlite3 .connect()	with mysql.connector .connect()	with cx_Oracle .connect()	with pyodbc .connect()
Query Placeholder	%s	?	%s	:name or ?	?
PEP 249 Compliant	Yes	Yes	Yes	Yes	Yes
Auto-commit Default	Off	Off	Off	Off	Off

Note on Query Placeholders: The placeholder syntax is the most common practical difference when switching between libraries. psycopg3 and mysql-connector-python both use %s, while sqlite3, pyodbc, and cx_Oracle (optionally) use ?. cx_Oracle also supports named placeholders (: name). All five libraries are PEP 249 DB-API 2.0 compliant, meaning the core patterns — connect, cursor, execute, fetch, commit — are consistent across all of them.